

Packet Tracer - Compare Layer 2 and Layer 3 Devices

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Objectives

Part 1: Compare Layer 2 and Layer 3 Switches

Part 2: Compare a Layer 3 Switch and a Router

Background / Scenario

In this activity, you will use various commands to examine three different switching topologies and compare the similarities and differences between the 2960 and 3650 switches. You will also compare the routing table of a 4321 router with that of a 3650 switch.

Note: Search the internet for more details about the *WS-C3650-24PS-L Layer 3 switch* and the *ISR 4321/K9 router*.

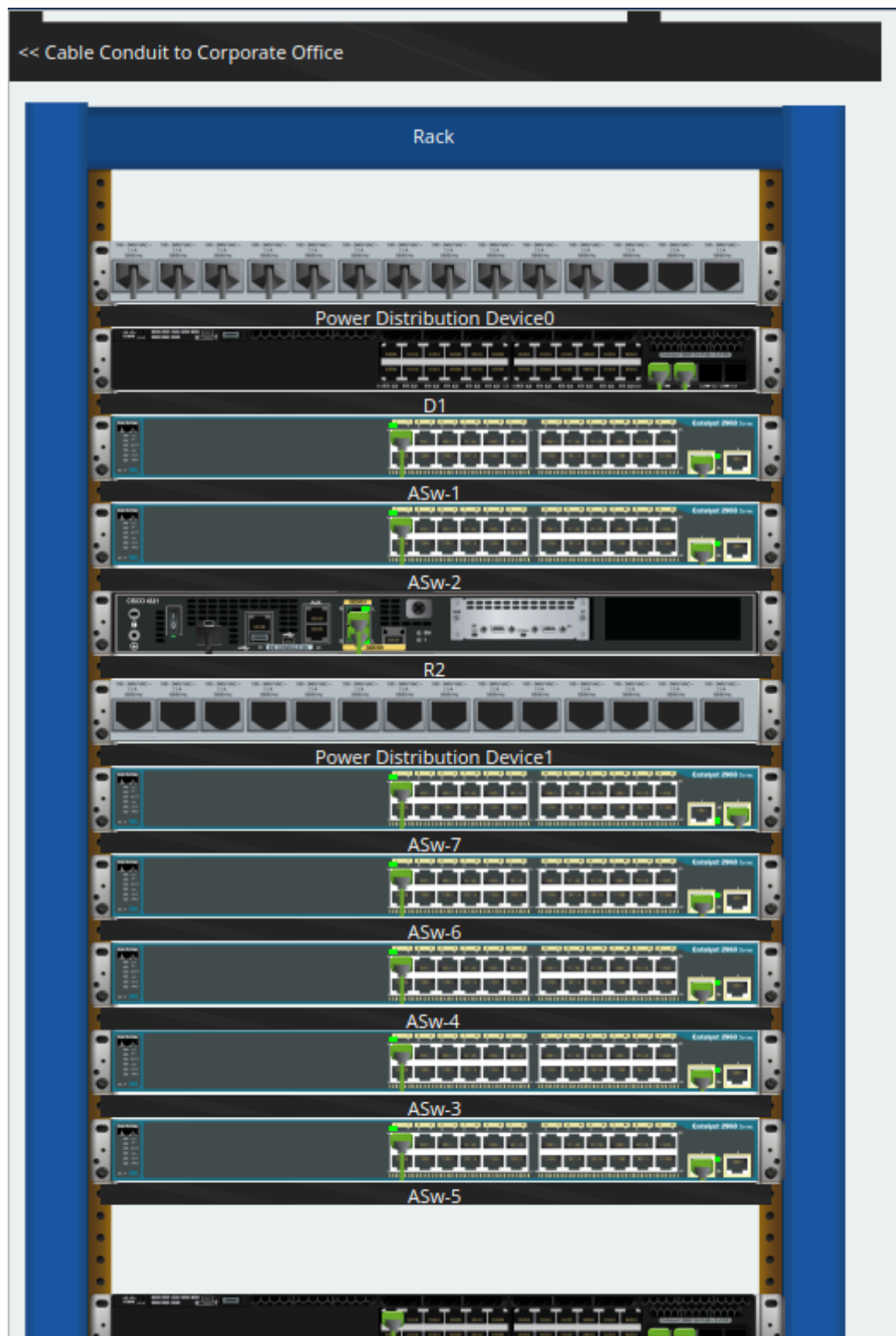
Instructions

Part 1: Compare Layer 2 and Layer 3 Switches

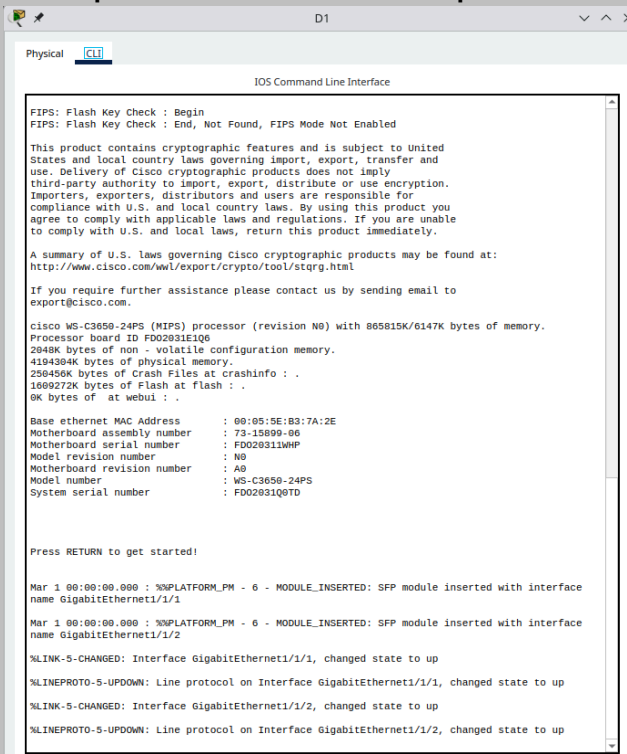
Step 1: Explore the Physical Workspace.

- a. In the Physical Workspace, click the **Home City** image. Click the **Corporate Office** image. Click the **Main Wiring Closet** image.
- b. In the Rack, locate the devices **D1** and **ASw-1**. Examine the physical aspects of **D1** and **ASw-1**. If you need to examine the devices more closely, click the device and select the Physical tab.

Each individual switch has how many physical switchports?



The 2960 has 26 ports. The 3650 has 24 fixed ports and four modular ports that accept SFP



```
Physical [CLI]
IOS Command Line Interface

FIPS: Flash Key Check : Begin
FIPS: Flash Key Check : End, Not Found, FIPS Mode Not Enabled

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wll/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco WS-C3650-24PS (MIPS) processor (revision N0) with 805815K/6147K bytes of memory.
Processor board ID FD02031E1Q6
2048K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
250456K bytes of Crash Files at crashinfo : .
1600272K bytes of Flash at flash : .
9K bytes of at webui : .

Base ethernet MAC Address      : 00:05:5E:B3:7A:2E
Motherboard assembly number    : 73-15899-06
Motherboard serial number      : FD020311WHP
Model revision number          : N0
Motherboard revision number    : A0
Model number                   : WS-C3650-24PS
System serial number           : FD02031Q0TD

Press RETURN to get started!

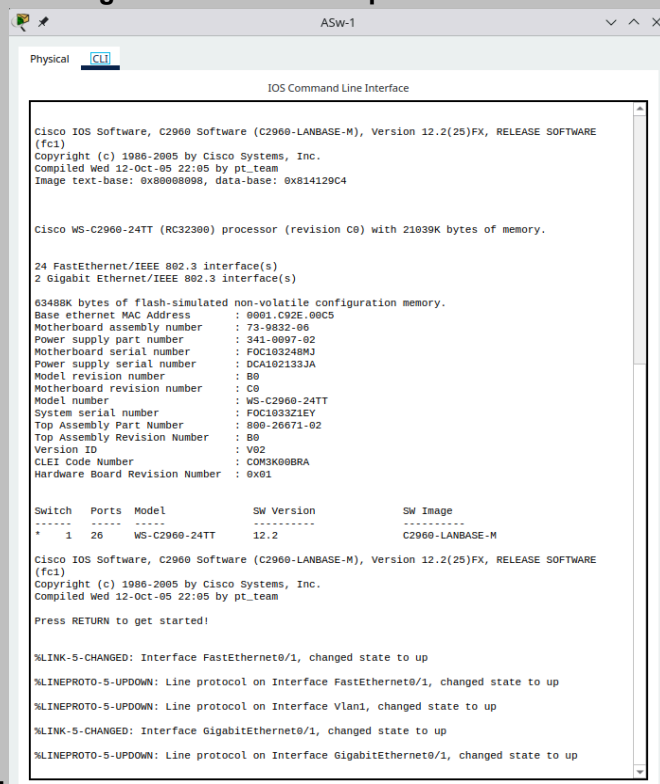
Mar 1 00:00:00.000 : %PLATFORM_PM - 6 - MODULE_INSERTED: SFP module inserted with interface
name GigabitEthernet1/1/1
Mar 1 00:00:00.000 : %PLATFORM_PM - 6 - MODULE_INSERTED: SFP module inserted with interface
name GigabitEthernet1/1/2

%LINK-5-CHANGED: Interface GigabitEthernet1/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/1/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet1/1/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/1/2, changed state to up
```

modules.

How many Fast Ethernet and Gigabit Ethernet switchports does each switch have?

The 2960 has 24 Fast Ethernet and 2 Gigabit Ethernet switchports. The 3650 has 24 fixed and 4



```
Physical [CLI]
ASw-1
IOS Command Line Interface

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE SOFTWARE
(fc1)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team
Image text-base: 0x80008008, data-base: 0x814129C4

Cisco WS-C2960-24TT (RC32380) processor (revision C0) with 21839K bytes of memory.

24 FastEthernet/IEEE 802.3 interface(s)
2 Gigabit Ethernet/IEEE 802.3 interface(s)

63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 0001.C02E.00C5
Motherboard assembly number    : 73-0532-06
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC103248MJ
Power supply serial number     : DCA102133JA
Model revision number          : B0
Motherboard revision number    : C0
Model number                   : WS-C2960-24TT
System serial number           : FOC10321EY
Top Assembly Part Number       : 800-26671-02
Top Assembly Revision Number   : B0
Version ID                     : V02
CLEI Code Number               : COM3K00BRA
Hardware Board Revision Number : 0x01

Switch  Ports  Model          SW Version      SW Image
-----  -
* 1    26    WS-C2960-24TT  12.2            C2960-LANBASE-M

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE SOFTWARE
(fc1)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

modular Gigabit Ethernet ports.

List the transmission speed of the Fast Ethernet and Gigabit Ethernet switchports on each switch.

```
D1#show running-config
Building configuration...

Current configuration : 1524 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname D1
.
```

```
ASw-1>enable
ASw-1#show running-config
Building configuration...

Current configuration : 1127 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ASw-1
```

The Fast Ethernet switchports support speeds of 10/100mb/s, and the Gigabit Ethernet switchports support speeds of up to 1000mb/s.

Are either of the two switches modular in design?

Yes, the 3650. อุปกรณ์ D1 (3650 - Layer 3 Switch) และ ASw-1 (2960 - Layer 2 Switch)

Step 2: In the CLI tab

- The switchports of a 3650 switch can be configured as Layer 3 interfaces by entering the **no switchport** command in interface configuration mode. This allows technicians to assign an IP address and subnet mask to the switchport in the same way that they are configured on a router interface.

What is the difference between a Layer 2 switch and a Layer 3 switch?

A Layer 2 switch makes forwarding decisions based on L2 (MAC) addresses. Switchports on Layer 3 switches can be configured as interfaces with IP addresses. The switches can also be configured with routing protocols like a router.

```
D1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet1/1/1
L       192.168.10.1/32 is directly connected, GigabitEthernet1/1/1
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet1/1/2
L       192.168.11.1/32 is directly connected, GigabitEthernet1/1/2
```

```
ASw-1#show ip route
^
% Invalid input detected at '^' marker.
```

What is the difference between a switch's physical interface and the VLAN interface?

A switch's physical interface is used to physically connect end devices to the network. A switched virtual interface (SVI or VLAN) is used to configure the switch with an IP address so that it can be managed remotely.

At which layers do 2960 and 3650 switches operate?

The 2960 operates at Layer 2, and the 3650 operates at Layers 2 and 3.

- b. Navigate to the CLI tab for both devices. Issue the **show run** command to examine the configurations of the **D1** and **ASw-1** switches.

Do you notice any differences between them?

Yes, D1's ports are all Gigabit Ethernet, while the 2960 has mostly Fast Ethernet ports and two Gigabit Ethernet ports that are intended for uplinks between switches. In addition, the D1 has different designations for its ports. The D1 uses the stack-module-port format. D1 has switch ports that are configured with the **no switchport** command and show an IP address and mask configured on the G1/1/1 and G1/1/2 ports. Also, D1 has IP routing enabled with the **ip routing** command

Try to display the routing table on D1 and ASw-1 using the **show ip route** command.

Why do you think the command does not work on **ASw-1** but works on **D1**?

It works on D1 because it functions on Layers 2 and 3, which allows it to function as a Layer 2 switch but at the same time, allows it to route packets and make forwarding decisions based on Layer 3 information (IP addresses) that conventional switches cannot. ASw-1 is a Layer 2 switch and therefore does not have a routing table.

Part 2: Compare a Layer 3 Switch and a Router

In the past, switches and routers have been separate and distinct devices. The term switch was set aside for hardware devices that function at Layer 2. Routers, on the other hand, are devices that make forwarding decisions based on Layer 3 information. They use routing protocols to share routing information and to communicate with other routers. Layer 3 switches, such as the 3650, can be configured to forward Layer 3 packets. Entering the **ip routing** command in global configuration mode allows Layer 3 switches to be

configured with routing protocols, which gives them some of the capabilities of a router. Although similar in some ways, Layer 3 switches are different from routers in many other aspects.

Step 1: Compare D1 and R1

- Open the Physical tab on D1 and R1.

Do you notice any similarities between the two? Do you notice any differences between the two?

They both have a console port, USB ports, and Gigabit Ethernet interfaces. Both R1 and D1 are modular, meaning different interfaces can be added. R1 has Serial and Asynchronous interfaces while D1 only has Ethernet interfaces. D1 can use copper or fiber-based Ethernet depending on the modules that are present and R1 can use various connection types also depending on the modules in use. D1 has many more Gigabit Ethernet ports than R1.

<pre>interface GigabitEthernet1/0/23 ! interface GigabitEthernet1/0/24 ! interface GigabitEthernet1/1/1 no switchport ip address 192.168.10.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/1/2 no switchport ip address 192.168.11.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/1/3 ! interface GigabitEthernet1/1/4 ! interface Vlan1 no ip address shutdown ! ip classless ! ip flow-export version 9 !</pre>	<pre>! interface GigabitEthernet0/0/0 ip address 192.168.10.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/0/1 ip address 192.168.11.1 255.255.255.0 duplex auto speed auto ! interface Serial0/1/0 no ip address clock rate 2000000 shutdown ! interface Serial0/1/1 no ip address clock rate 2000000 shutdown ! interface Vlan1 no ip address shutdown ! ip classless ! ip flow-export version 9 !</pre>
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- In the CLI tab, issue the **show run** command and examine the configurations of R1 and D1.

What differences do you see between the two?

R1 and D1 have the same IP addresses configured on them but on different interfaces.

<pre>D1# D1#show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is not set 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.10.0/24 is directly connected, GigabitEthernet1/1/1 L 192.168.10.1/32 is directly connected, GigabitEthernet1/1/1 C 192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.11.0/24 is directly connected, GigabitEthernet1/1/2 L 192.168.11.1/32 is directly connected, GigabitEthernet1/1/2</pre>	<pre>R1# R1#show ip route Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is not set 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.10.0/24 is directly connected, GigabitEthernet0/0/0 L 192.168.10.1/32 is directly connected, GigabitEthernet0/0/0 C 192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks C 192.168.11.0/24 is directly connected, GigabitEthernet0/0/1 L 192.168.11.1/32 is directly connected, GigabitEthernet0/0/1</pre>
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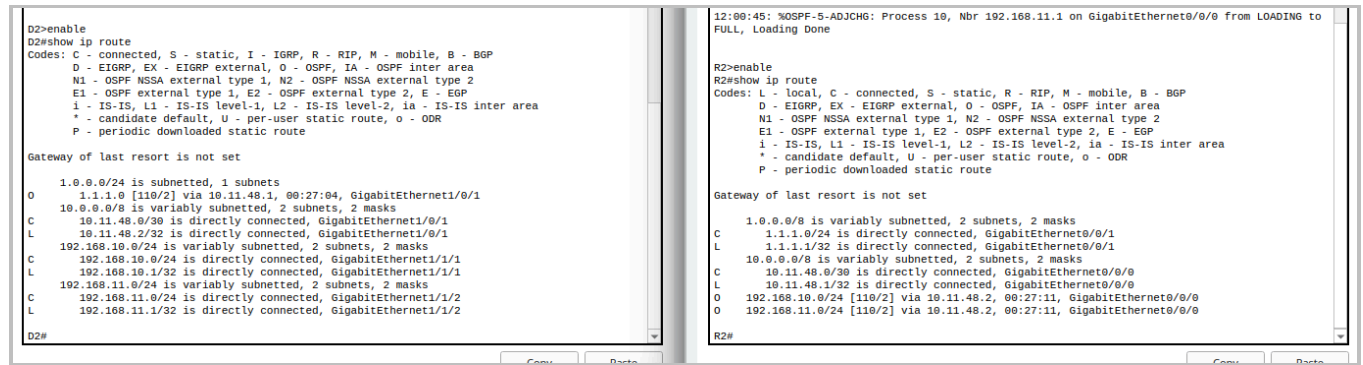
Which command allows configuration of D1 with an IP address on one of its physical interfaces?

The no switchport command.

- c. Use the **show ip route** command on both devices.

Do you see any similarities or differences between the two tables?

The codes are the same except the router has an L code for local. This is a link that is configured on the physical interface of R1. The switch routing table does not have this code. Both devices display the same networks in their routing tables.



Step 2: Compare R2 and D2

- a. Now, analyze the routing table of R2 and D2. Click both devices in the Rack. In the CLI tab, issue the **show ip route** command.

What is present now that was not present in the configuration of R1 and D1?

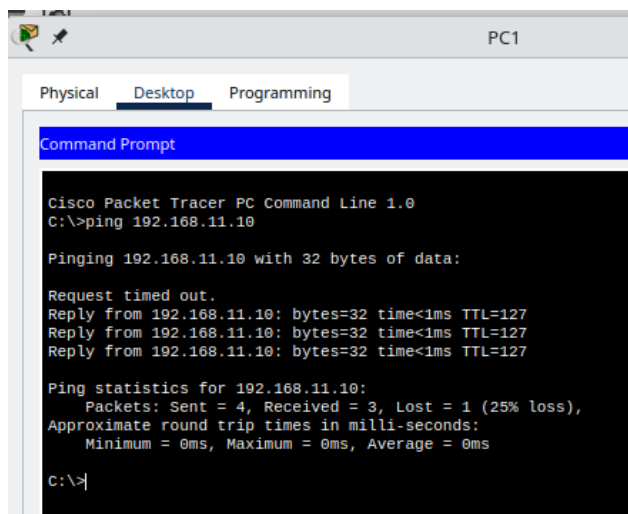
They both have OSPF configured and both are learning networks from one another.

Which network is in the routing table of D2 that was learned from R2?

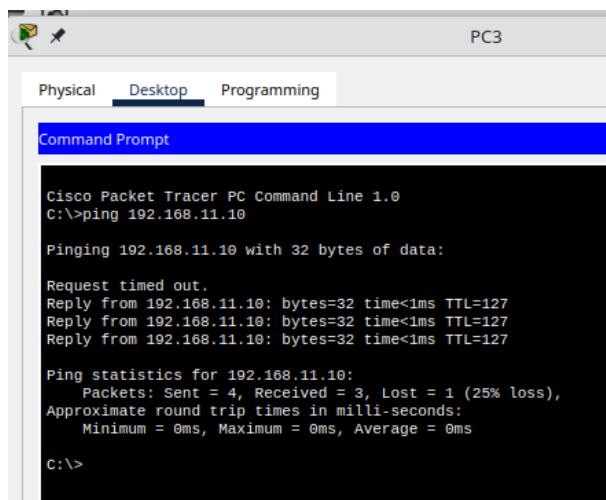
The 1.1.1.0/24 network was learned from R2.

- b. Click the **Back Level** icon (Alt + Left) to leave the Main Wiring Closet. Verify that each topology has full connectivity by completing the following tests:

Ping from **PC1** to **PC2**



Ping from **PC3** to **PC4**



```
PC3
Physical Desktop Programming
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.10

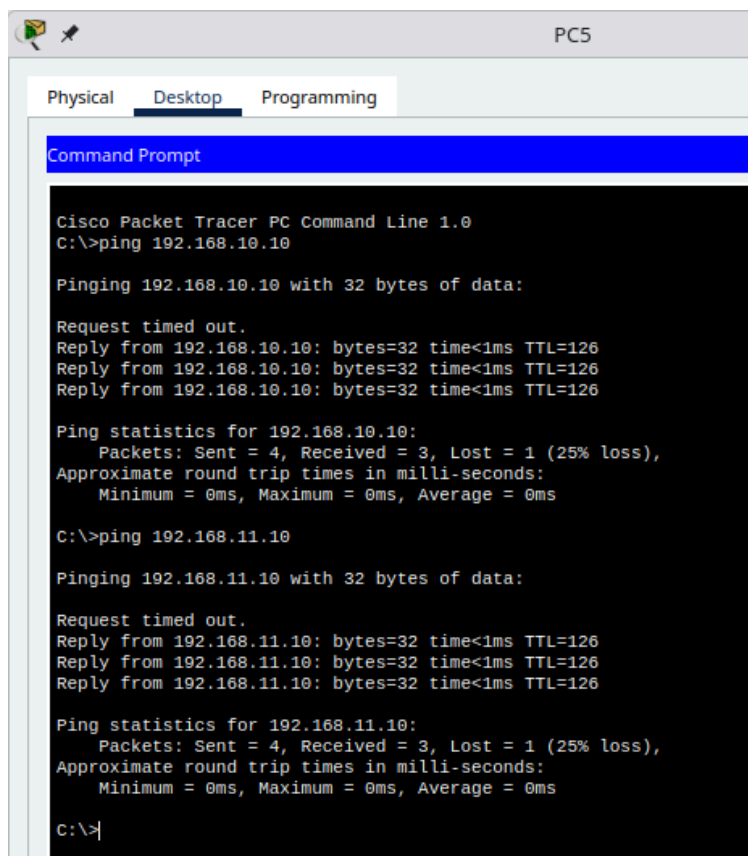
Pinging 192.168.11.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Ping from **PC5** to **PC6** and **PC7**



```
PC5
Physical Desktop Programming
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.11.10

Pinging 192.168.11.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

In all three examples, each PC is on a different network.

Note: You may need to move the PCs to see them.

Which device is used to provide communication between networks?

Router or multilayer switch.

Why were we able to ping across networks without there being a router?

A multilayer switch can route between networks as long as it is configured with an IP address and has IP routing enabled. IP routing must also be enabled if you plan to run routing protocols such as OSPF on the switch. The `no switchport` command must be enabled on the interface in order to assign an IP address and subnet mask on the switch's physical interface.

Bonus question: We say that routers are Layer 3 devices and conventional (non-Layer 3) switches are Layer 2 devices. However, we can assign an IP address to a management (SVI) interface of a Layer 2 switch. How is this possible if switches are Layer 2 devices?

Managed Layer 2 switches, such as the Cisco Catalyst 2960, have an embedded server that can be reached over Layer 3. The server enables Telnet, SSH, or HTTP access to the switch from across the network so that the switch can be remotely managed and configured. It is useful to think of this functionality as separate from the data forwarding function of the switch, which exists at Layer 2.